# TikTok Claims Classification Project

Random Forest Classifier — Executive Summary

## ■ ISSUE / PROBLEM

TikTok receives millions of user-reported videos daily — far too many for human moderators to review individually. Analysis shows that when authors violate the platform's terms of service, they are much more likely to be posting claims than opinions. A machine learning model is needed to triage reports efficiently.

**Ethical considerations:**

- False negatives (claims missed) are far more harmful than false positives
- Evaluation metric: RECALL — minimize missed violations
- Human review remains essential — model triages, not decides
- Transparency and appeal process recommended for flagged creators

## ■ RESPONSE

A Random Forest classifier was built to predict whether a TikTok video is a claim or an opinion. Two models were developed and compared — Random Forest and XGBoost — with hyperparameters tuned via 5-fold GridSearchCV optimizing for recall.

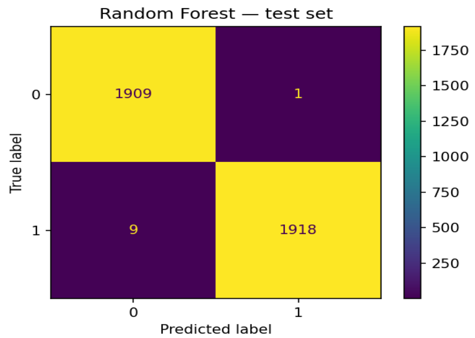
**Modeling workflow:**

- 60/20/20 train / validate / test split
- GridSearchCV with 5-fold cross-validation
- Model selection on validation set
- Final evaluation on held-out test set
- Champion: Random Forest (higher recall than XGBoost)

## ■ MODEL PERFORMANCE

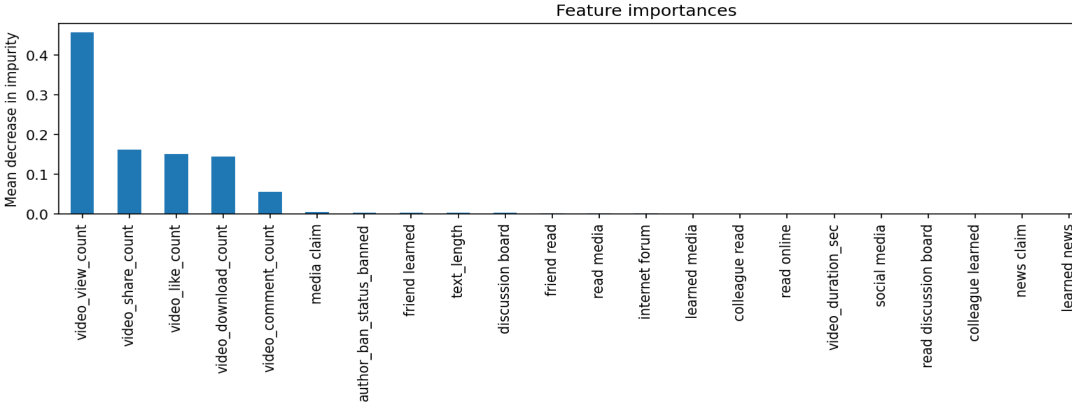
**Model comparison**

| Model | Dataset | Recall | Precision | Accuracy |
|---|---|---|---|---|
| Random Forest | Cross-val (train) | 99.5% | 99.98% | — |
| Random Forest | Validation | 99.6% | 100% | 99.8% |
| Random Forest ■ Champion | Test | 99.5% | 99.9% | 99.7% |
| XGBoost | Validation | 99.1% | 99.9% | 99.5% |




*Champion model test set confusion matrix: only 10 misclassifications out of 3,837 samples*

## ■ FEATURE IMPORTANCES




The most predictive features are all engagement metrics: **video_view_count (45.7%)** is by far the most important, followed by share count (16.2%), like count (15.0%), and download count (14.5%). Text features, verified status, and ban status contribute very little. The model classifies videos primarily based on how much engagement they receive — claim videos consistently generate dramatically higher engagement than opinion videos.

## ■ KEY INSIGHTS

The Random Forest classifier achieved near-perfect performance across all evaluation sets. The two key insights from this final model are:

**Engagement is the strongest signal**

Claim videos receive dramatically higher engagement than opinion videos. The model learned that view count alone predicts claim status with ~45% of total feature importance. This suggests the TikTok community actively amplifies unverified claims through sharing and viewing behavior.

**Model is ready for production**

With 99.7% test accuracy and 99.5% recall, this model is recommended for deployment. Classified claim videos should be ranked by report count for human review priority.

**Recommended next steps**

- Deploy model to triage reported videos — claims go to review queue ranked by report count
- Introduce human review checkpoint for top-flagged clips daily
- Collect number of times each video is reported as an additional feature
- Monitor model performance quarterly and retrain as content patterns evolve
- Explore NLP models using full transcription text for potential further improvement